ELECTRIC VEHICLE TECHNOLOGY

TASK 2: BATTERY ANALYSIS REPORT

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1. STUDY OF LITHIUM-ION BATTERY PACKS IN EVS

Lithium-ion (Li-ion) batteries are the dominant energy storage technology in modern electric vehicles (EVs) due to their superior energy density, light weight, and long cycle life. Understanding battery architecture and chemistry selection is fundamental to mechanical and electrical EV design integration.

1. Battery Cell

The smallest functional unit (Cylindrical 21700, Prismatic, or Pouch). Generates standard nominal voltage (typically 3.2V - 3.7V).

2. Battery Module

Multiple cells connected in series/parallel housed in a protective frame with thermal cooling plates and voltage/temperature sensors.

3. Battery Pack

Final integrated assembly combining modules, central BMS, high-voltage contactors, thermal management circuits, and rugged enclosure.

2. BATTERY CAPACITY & ENERGY CALCULATIONS

To dimension an electric vehicle battery pack, engineers calculate total energy demand based on target driving range, average energy efficiency, and safety buffers for battery lifespan longevity.

Design Example & Mathematical Sizing:

Given Specifications: Target Vehicle Range = **150 km** | Average Energy Consumption = **120 Wh/km** | Pack Voltage = **400 V** | Max Depth of Discharge (DoD) = **80% (0.8)**

1. Required Net Energy (E_{net}) = Range × Energy Consumption	150 km × 120 Wh/km = **18,000 Wh (18 kWh)**
2. Total Pack Capacity (E_{total}) = E_{net} / DoD	18 kWh / 0.8 = **22.5 kWh**
3. Ah Capacity Requirement = (E_{total} × 1000) / Nominal Voltage	**22,500 Wh / 400 V = 56.25 Ah**

3. COMPARISON OF EV BATTERY CONFIGURATIONS

Selecting the optimal battery chemistry involves balancing trade-offs between energy density, safety, lifespan, and cost depending on the intended vehicle application.

Parameter / Feature	Lithium Nickel Manganese Cobalt (NMC)	Lithium Iron Phosphate (LFP)
Specific Energy Density	High (160 – 250 Wh/kg)	Moderate (100 – 160 Wh/kg)
Thermal Stability	Moderate (Thermal runaway onset ~210°C)	Excellent (Thermal runaway onset ~270°C)
Cycle Life (100% DoD)	1,500 – 2,500 cycles	3,000 – 6,000+ cycles
Raw Material Cost	Higher (Contains expensive Cobalt & Nickel)	Lower (Abundant Iron & Phosphate)
Primary Application	Long-range passenger cars, high-performance EVs	Standard-range EVs, heavy vehicles, commercial buses

4. IMPORTANCE OF BATTERY MANAGEMENT SYSTEMS (BMS)

The Battery Management System (BMS) is an embedded electronic system that acts as the core safety regulator and operational brain of an electric vehicle's high-voltage pack.

State Estimation (SoC & SoH)

Accurately estimates Stateof Charge (remaining battery range %) and State of Health (overall pack degradation over time) using complex algorithm models.

Active & Passive Cell Balancing

Monitors individual cellvoltages toequalize charge levels across all cells, preventing single cell overcharging or under-voltage limits during operation.

Thermal Management Control

Monitors packtemperaturenodes and activates liquid cooling or resistance heating loops to keep cells within their optimal operating window (20°C to 40°C).

Safety & Fault Protection

Instantly triggershigh-voltage contactor disconnects during over-current, short circuits, ground faults, or thermal runway conditions to ensure passenger safety.

Mechanical Engineer's Summary

Asamechanicalengineeringstudent, evaluating battery pack design requires a multi-disciplinary approach considering structural integration, heat dissipation, structural containment, and electrical efficiency. Selecting between NMC and LFP configurations directly influences chassis balance, total curb weight, and structural crash safety frames.